

Service Publication

Diesel Engine
12V 396 TB84

Description and
Operation Manual
M010855/00E



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M010855/00E Description and Operation Manual 12V 396 TB 84	Postcard MTU Friedrichshafen GmbH Department SCT 88040 Friedrichshafen GERMANY
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This manual is intended to inform you about your MTU diesel engine and to provide assistance for operation, servicing and maintenance. It is divided up as follows:

Part A PRODUCT SUMMARY

Group A - Engine layout and technical data

Part B DESCRIPTION

Groups B00 to B88 - Descriptions of components and systems

Part C OPERATING INSTRUCTIONS

Group C - Schedule of services

Part D MAINTENANCE

Group D - Time schedule, job schedule

Part E TROUBLESHOOTING

Group E - Faults, causes and remedies

Part F DRAWINGS

Group F - Technical section drawings

Part G TASK DESCRIPTION

Groups G00 to G88 - Inspection and maintenance operations.

Part K PRESERVATION

Group K - Engine preservation

Part Z APPENDIX

Group Z - Engine installation and general arrangement drawings, system diagrams, Engine Acceptance Test Records and subcontractors' documents.

Subcontractor designations and specifications relating to engine-mounted ancillaries depict the engine configuration current at the time of publication of this manual. These specifications must not be used for parts ordering under any circumstances.

Diesel Engine

Part A
Product Summary

The product summary gives details of the engine and contains technical data such as engine power, general engine specifications, engine weight, oil and coolant capacities, settings and operational data.

CONTENTS

A010	Engine summary
A020	Engine side and cylinder designations
A030	General specifications
A040	Main engine dimensions, engine weight
A050	Operational data

ENGINE LAYOUTKey to Engine Model Designation 12 V 396 TB 84

- 12 = number of cylinders
- V = Vee configuration
- 396 = engine series - 100 times displacement of one cylinder in liters
- T = exhaust turbocharging
- B = external charge air-to-liquid cooling, with piston cooling
- 8 = engine for marine propulsion
- 4 = design digit

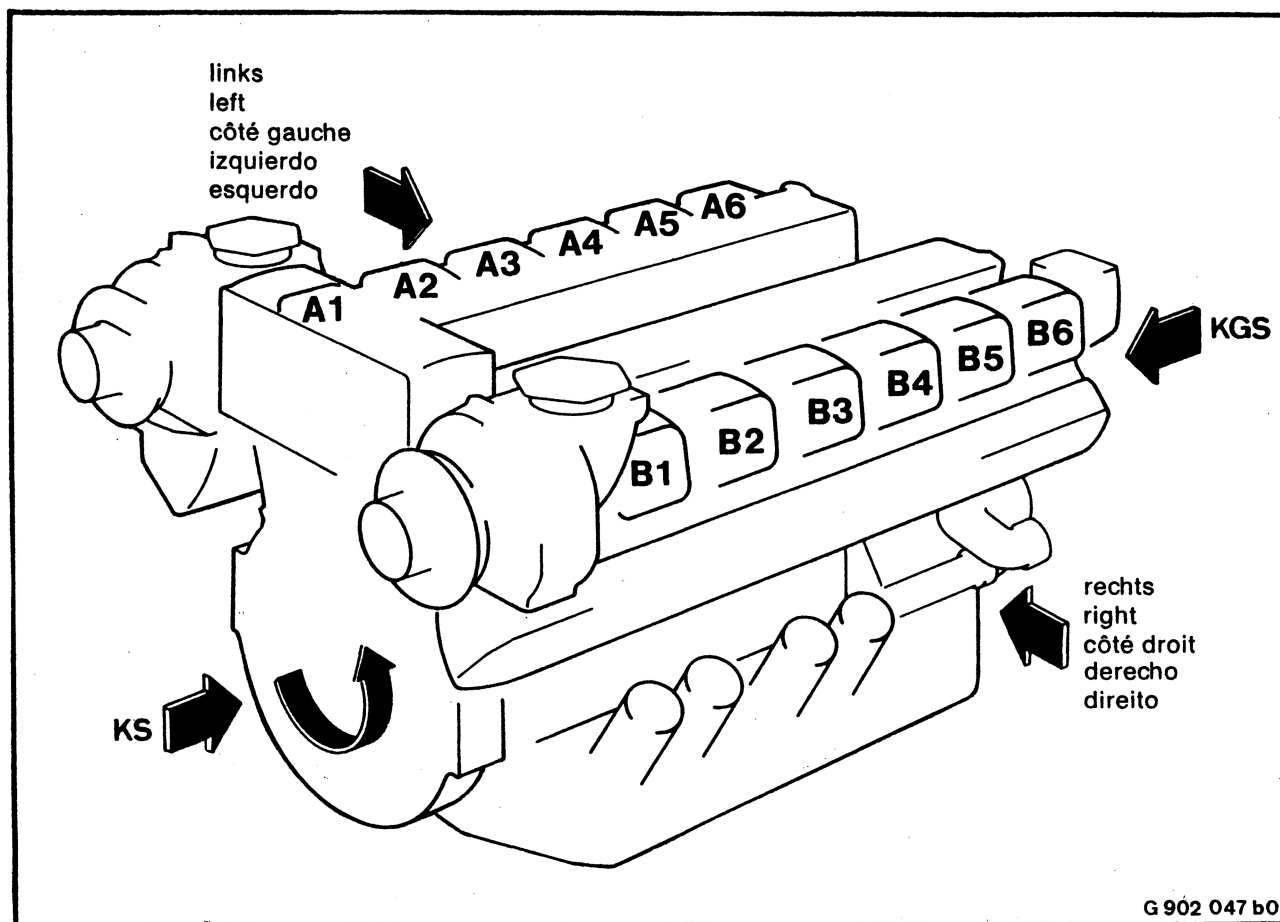
Engine Power

Continuous power 1540 kW at 1940 rpm

Overload power 1680 kW at 2000 rpm
permissible for 2 hours within 12 operating hours

Reference conditions

- intake air temperature 45°C
- raw water temperature 32°C
- barometric pressure 1000 mbar

ENGINE SIDE AND CYLINDER DESIGNATIONS

Engine sides are always designated as viewed from the flywheel end.

The left bank of cylinders is marked "A" and the right bank "B" (according to DIN standard 6265).

The cylinders of each bank are numbered consecutively, starting with No. 1 at the flywheel end.

Numbering of other engine components is also from the flywheel end, starting with No. 1.

Designations and abbreviations

Flywheel end = KS (formerly HKS)

Timing end = KGS (formerly GKS)

Left side (formerly A side)

Right side (formerly B side)

GENERAL SPECIFICATIONS

Operating method	four-stroke cycle, single-acting
Combustion method	direct injection
Mode of supercharging	exhaust turbocharging sequential turbocharging (single-stage)
Cooling method	liquid-cooling
Cylinder arrangement	90° Vee
Bore	165 mm
Stroke	185 mm
Displacement, cylinder	3.96 liters
Number of cylinders	12
Displacement, total	47.5 liters
Compression ratio	11 : 1
Direction of rotation (viewed from flywheel end)	c.c.w. (non-reversible)
Firing order	A1 A5 A3 A6 A2 A4 B5 B3 B6 B2 B4 B1
Fuel injection pressure:	
Setting pressure with new spring	280 to 288 bar
Specified pressure with run-in spring	270 to 278 bar
Minimum pressure with run-in spring	250 bar
Final compression pressure at 120 rpm 736 Torr and engine coolant at 5°C	19 to 23 bar
Break-away torque at engine coolant temperature of 40°C, allowance for acceleration included	600 Nm
Cranking torque at engine coolant temperature of 40°C	560 Nm
Firing speed at engine coolant temperature of 40°C	approx. 120 rpm
Mean piston speed	
at 1975 rpm	12.17 m/s
at 2040 rpm	12.58 m/s
at 2100 rpm	12.95 m/s

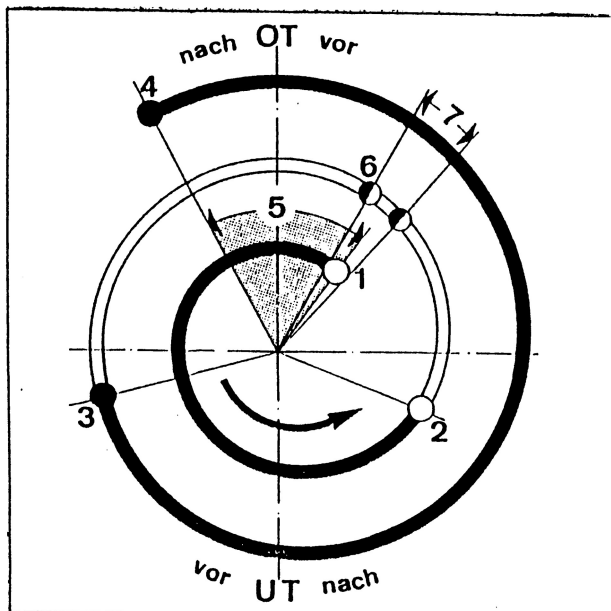
Valve clearance, cold engine

Inlet 0.30 mm
Exhaust 0.45 mm

Timing

(valve clearance adjusted)

(crankshaft degrees)



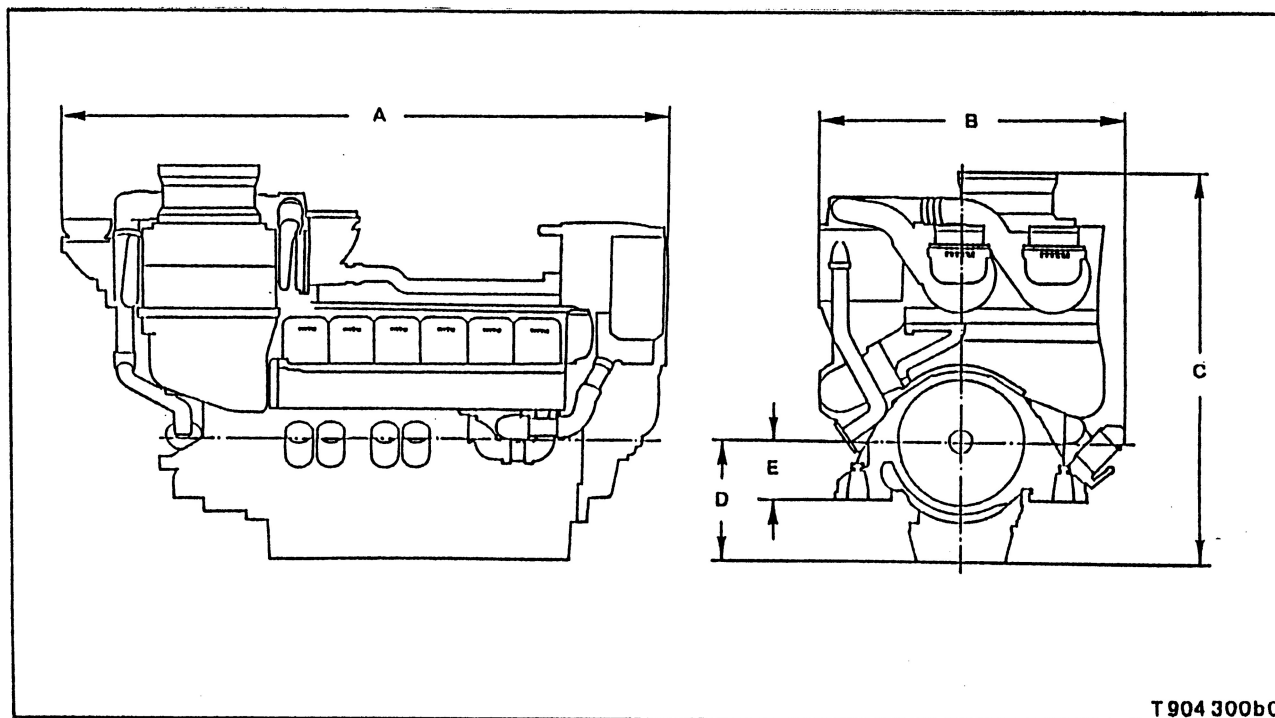
TA920 023

nach = after
vor = before
OT = TDC
UT = BDC

- | | |
|--|----------------|
| 1 Inlet valve opens | 36° before TDC |
| 2 Inlet valve closes | 68° after BDC |
| 3 Exhaust valve opens | 75° before BDC |
| 4 Exhaust valve closes | 28° after TDC |
| 5 Overlap | 64° |
| 6 Start of injection pump delivery see Engine Acceptance Test Record | |
| 7 Adjustment range of injection timer | max. 14° |

MAIN ENGINE DIMENSIONS, ENGINE WEIGHT

Main Engine Dimensions



T 904 300b0

A = Overall length	2992 mm
B = Overall width	1420 mm
C = Overall height	1881 mm
D = Crankshaft height from oil pan	535 mm
E = Crankshaft height from engine mount	230 mm

Engine Weight, Coolant and Oil Capacities

Engine dry weight	approx. 5145 kg
Coolant capacity including engine-mounted pipework	approx. 180 liters
Raw water capacity including engine-mounted pipework	approx. 45 liters
Oil capacity, engine	
- up to lower mark on dipstick	approx. 95 liters
- up to upper mark on dipstick	approx. 135 liters
Total capacity	
- for initial filling	approx. 180 liters
- for oil change	approx. 155 liters